

A LIGHTWEIGHT FRAMEWORK FOR PRIVACY-PRESERVING BEHAVIORAL AUTHENTICATION

Balancing Recognition Accuracy and System Latency

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Presentation Outline

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2. **Literature Review:** Behavioral Authentication Evolution
3. **Problem and Significance:** Technical Dichotomy
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Research Context and Motivation

- ▶ **Limits of Static MFA:** Conventional security requires repetitive, intrusive manual entry for small tasks like messaging or authorizing transactions.
- ▶ **Transition to Behavioral Biometrics:** Keystroke and mouse dynamics offer a robust alternative by verifying users through unique biomechanical patterns.
- ▶ **Continuous Passive Verification:** Replaces one-time logins with background monitoring to ensure the authorized user remains in control without workflow interruption.
- ▶ **The Privacy Conundrum:** Unlike passwords, biometric traits are inherent and immutable; a raw data breach results in permanent identity loss.
- ▶ **Data Liability:** Storing raw templates in central databases creates high-risk vulnerabilities and potential digital identity theft.

Evolution of Behavioral Authentication

Reference Study	Key Contribution to This Project	Identified Gap / Limitation
Gaines et al. (1980)	Foundational Theory: Established typing rhythms (dwell/flight time) as unique for verification.	Relied on static, fixed-text passwords; insufficient for continuous authentication.
Mondal & Bours (2017)	Multimodal Fusion: Proved combining Keystroke and Mouse dynamics significantly reduces EER.	High-dimensional vectors created by concatenation slow down real-time processing.
Kim et al. (2018)	Feature Engineering: Introduced "user-adaptive" features to improve personalized accuracy.	Lacked template protection or encryption to secure stored biometric data.
Kim et al. (2024)	Deep SVDD Validation: Outperformed traditional models (6.7% EER) via time-series to image encoding.	Restricted to mobile PINs; lacked multimodal fusion or cryptographic encryption (HE).
Kiyani et al. (2020)	Continuous Authentication: Validated RNNs for ongoing user verification beyond just login.	Did not address high latency introduced when adding encryption to continuous streams.
Rahman et al. (2021)	Scalability Metrics: Provided framework for measuring error rates as user database size increases.	Addressed scalability of accuracy but not the scalability of secure template storage.

Table: Summary of Key Related Studies

The Technical Dichotomy and Significance

- ▶ **The Primary Problem:** An unresolved trade-off exists between system latency, user privacy, and template revocability.
- ▶ **Efficiency Gap:** Cryptographic tools like Homomorphic Encryption (HE) are too computationally expensive for real-time edge monitoring.
- ▶ **Privacy Gap:** High-accuracy Deep Learning models often require storing original behavioral features, making them prone to reverse engineering.
- ▶ **Local Template Recoverability:** This research enables "cancelable" biometrics where compromised data is reset by regenerating a **Keyed-JL Projection Matrix** locally.
- ▶ **Significance:** Provides a mechanism to re-issue a fresh behavioral identity on the local computer without requiring the user to change their natural habits.

The Technical Dichotomy

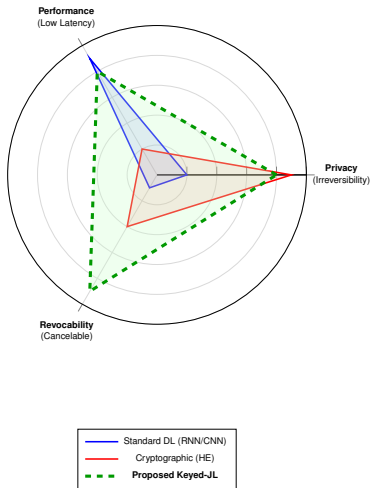


Figure: Technical Trade-offs Comparison

Formal Research Gap Definition

The research gap is formally defined as the lack of a unified behavioral authentication framework that provides **provable non-invertibility** without sacrificing the **sub-second latency** required for continuous monitoring.

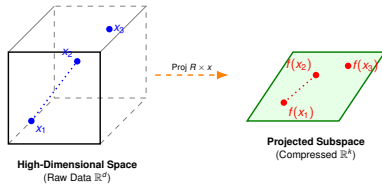
While existing studies achieve high accuracy using **DL! (DL!)**, they fail to provide a mechanism for template revocability, meaning a compromised biometric profile is permanently lost. Conversely, cryptographic methods like **HE! (HE!)** that provide privacy are too computationally expensive for real-time use on standard hardware.

- **DL Models:** Lack template protection and revocability.
- **HE Methods:** High computational overhead and latency.

Research Aim and Technical Objectives

1. **To design a privacy-preserving feature transformation pipeline:** Develop a mechanism using Keyed-JL! (JL!) Projections that transforms high-dimensional multimodal features into a lower-dimensional subspace, ensuring templates are mathematically irreversible and revocable through key renewal.
2. **To implement a multimodal behavioral fusion layer:** Combine independent keystroke dynamics (dwell/flight time) and mouse dynamics (velocity/curvature) into coordinated behavioral units (KMT! (KMT!)) to raise the entropy of the user profile.
3. **To develop a lightweight server-side anomaly detection model:** Implement a **Deep SVDD! (Deep SVDD!)** classifier to distinguish users by learning a compact hypersphere boundary around projected privacy-preserving embeddings.
4. **To evaluate the trade-off between privacy, accuracy, and latency:** Quantify the impact of projection dimension k on authentication performance (**EER! (EER!)**), system latency (target $< 200\text{ms}$), and resistance to recovery attacks.

Theoretical and Technical Foundations



JL Lemma Guarantee:
 $\|f(x_1) - f(x_2)\| \approx \|x_1 - x_2\|$

Figure: JL Projection: Dimension reduction.

- **Keyed-JL:** Uses secret key for mathematical irreversibility and revocability.

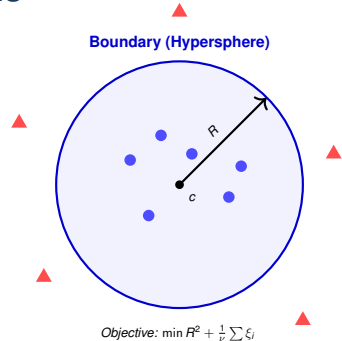
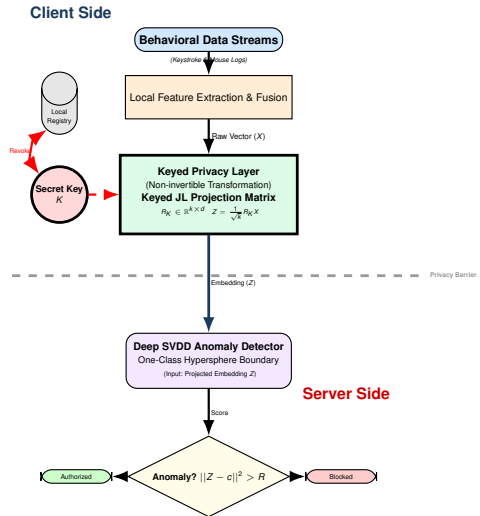


Figure: Deep SVDD: Hypersphere boundary.

- **Anomaly Detection:** Rejects any input falling outside the learned radius R .

Detailed System Architecture

- **Privacy-by-Design:** Raw data never leaves the local device, ensuring data minimization.
- **Keyed Transformation:** Uses a secret key K to generate non-invertible templates.
- **Privacy Barrier:** Only projected embeddings (Z) cross the network to the server.
- **One-Class Engine:** Deep SVDD detects anomalies without requiring imposter data.
- **Template Revocability:** Compromised identities can be "reset" by local key renewal.



Multimodal Datasets and Evaluation Framework

Hybrid Multi-Source Datasets

- ▶ **SU-AIS BB-MAS**: Primary source for Deep SVDD training ($N = 117$ subjects).
- ▶ **CMU Benchmark**: Baseline control for keystroke dynamics validation.
- ▶ **Edge Hill KMT**: Critical for multimodal fusion and trajectory analysis.

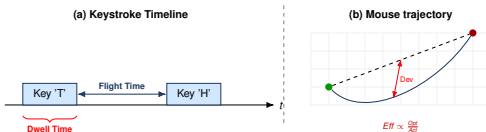


Figure: Visual Representation of Biometric Features.

Authentication Metrics

- ▶ **EER**: Point where False Acceptance and False Rejection rates are equal.
- ▶ **FAR/FRR**: Standard frequency measures for unauthorized access and incorrect denials.

Privacy and Efficiency

- ▶ **Resistance to Attack**: Quantified by Reconstruction Error (MSE_{recon}).
- ▶ **Revocability**: Validation of uncorrelated templates via key renewal.
- ▶ **Latency** (L_{inf}): Total time from projection to decision ($< 200ms$ target).

Anticipated Outcomes and Novel Contributions

Expected Technical Outcomes

- ▶ **Privacy Protection:** Mathematical proof that raw keystroke and mouse data cannot be reconstructed from templates.
- ▶ **Template Revocability:** Validation that compromised identities can be "reset" and re-issued via key renewal.
- ▶ **High-Efficiency Detection:** Empirical evidence of real-time anomaly detection suitable for edge devices.
- ▶ **Competitive Accuracy:** Authentication performance comparable to systems relying on raw templates.

Novel Contributions

- ▶ **Unified Architecture:** Integration of Keyed-JL random projections with Deep SVDD for behavioral biometrics.
- ▶ **Lightweight Alternative:** Provision of a computationally efficient alternative to heavy cryptographic methods.
- ▶ **Trade-off Characterization:** Analysis of the privacy-latency-accuracy trade-off using the JL Lemma.
- ▶ **Edge-Ready Design:** A framework specifically architected for resource-constrained IoT environments.

Research Assumptions and Technical Constraints

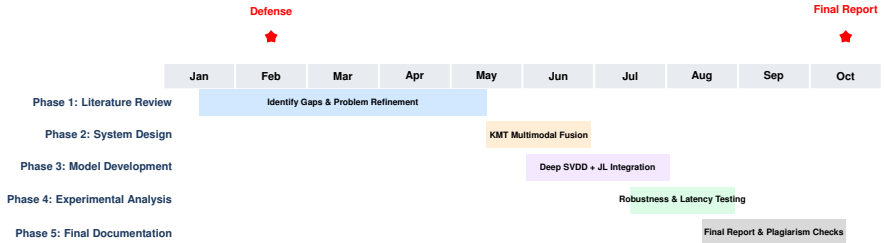
Foundational Assumptions

- ▶ **Behavioral Stability:** Primary user patterns remain relatively stable over the study duration.
- ▶ **Secure Enrollment:** Initial user enrollment occurs in a verified, non-adversarial environment.
- ▶ **Key Integrity:** The secret projection key is stored securely on the client-side device.
- ▶ **Hardware Sufficiency:** Target devices possess sufficient memory to perform linear JL-projections.

Technical Constraints

- ▶ **Modality Scope:** Limited to desktop inputs (KMT); does not include mobile-specific sensors.
- ▶ **Data Bounds:** Evaluation is restricted to the diversity present in the selected benchmark datasets.
- ▶ **Latency-Privacy Trade-off:** System must operate within a strict real-time bound of $< 200\text{ms}$.
- ▶ **Environmental Noise:** Results may vary based on differences in physical hardware peripherals.

Project Timeline and Detailed Task Breakdown



Phases 1 – 3: Development

- **Review:** Technical dichotomy between privacy and performance.
- **Design:** Multimodal feature fusion for coordinated behavioral units.
- **Dev:** Optimize ReLU activation and SVDD hypersphere boundaries.

Phases 4 – 5: Validation

- **Evaluation:** Verify privacy robustness against template inversion attacks.
- **Analysis:** Formally characterize the privacy-latency trade-off.
- **Reporting:** Prepare final report and plagiarism checks.

Research Summary and Future Outlook

Project Summary

- ▶ **Addressing the Gap:** Developed a unified framework to resolve the conflict between privacy, system latency, and template protection.
- ▶ **Technical Synergy:** Successfully integrated Keyed-JL Projections for security with Deep SVDD for efficient anomaly detection.
- ▶ **Multimodal Advantage:** Leveraged fused keystroke and mouse dynamics to enhance user profile entropy and recognition accuracy.

Closing Statement

This research bridges the gap between privacy theory and practical real-time authentication, contributing toward more secure and edge-deployable digital identity systems.

Thank You!

Questions?

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